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### **ELECTRONIC DEVICES INCLUDING DEVICE PACKAGES HAVING A PORT HOLE WITH A SPARK ARREST SHOULDER AND RELATED METHODS**

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#### **Abstract**

An electronic device may include a device that communicates with the environment of the user, which may include acoustic, radio, optical, etc., information. The device is mounted on a first side of a multi-layer substrate in a device package. To provide communicative access to the environment, the device is mounted on a first opening of a port hole that extends through the multi-layer substrate. The port hole includes a second opening in a first substrate layer on a second side of the multi-layer substrate, where the second opening is wider than a third opening in a second substrate layer, to form a shoulder region that is exposed through the second opening and on which a conductive layer coupled to a reference voltage can provide a spark arrestor to protect the device from damage due to electric sparks entering from the environment.

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## Background/Summary

### BACKGROUND

#### I. Field of the Disclosure

[0001] The technology of the disclosure relates generally to reducing spark-related damage to electronic devices and, more particularly, to spark arrestors in user devices.

#### II. Background

[0002] Electronic devices may include one or more internal devices for communicating with the environment, such as microphones, speakers, pressure sensors, and optical sensors. To access the environment, the electronic device may include a port hole providing a path between the internal device and the environment. When these devices are frequently handled by users, they may develop electrostatic charge differences between the user and/or nearby objects. When such charge differences exceed a threshold, a discharge may occur in either direction in the form of a spark. The internal devices of an electronic device have an electrical potential that can make them an attractive destination for such discharges, especially because such internal devices are exposed to the environment through the port holes needed for their operation. Consequently, such devices are highly susceptible to damage from sparks. A burst of charge from a spark can damage electrical components, and a sudden expansion of gas caused by localized heating from the spark can also damage mechanical components.

### SUMMARY

[0003] Aspects disclosed in the detailed description include electronic devices including device packages having a port hole with a spark arrest shoulder. Related methods of forming a device package, including a spark arrest shoulder, are also disclosed. The electronic device may include a device that receives information communicated in signals from the environment of the user outside of the device, which may include acoustic signals, radio signals, or optical signals, for example. The device is mounted on a first side of a multi-layer substrate as part of a device package. To provide communicative access to the environment, the device is mounted on a first opening of a port hole that extends through the multi-layer substrate from a second side, opposite to the first side, so that a signal from the environment may pass through the port hole to reach the device. In an exemplary aspect, the package includes a spark arrestor disposed inside the port hole and coupled to a reference voltage. The spark arrestor is provided to attract sparks to protect the device from damage due to electric sparks entering from the environment. In some examples, the port hole includes a second opening in a first substrate layer on the second side of the multi-layer substrate. In such examples, the second opening may be wider than a third opening in a second substrate layer, forming a shoulder region inside the port hole that is exposed through the second opening. The spark arrestor may include a conductive layer on the shoulder region.

[0004] In this regard, in one exemplary aspect, a package is disclosed. The package includes a multi-layer substrate, a port hole through the multi-layer substrate and including a first opening on a first side of the multi-layer substrate and a second opening on a second side of the multi-layer substrate, opposite to the first side. The package further includes a device disposed on the first opening of the port hole and having a first voltage and a spark arrestor, including a first conductive layer of the multi-layer substrate disposed inside the port hole and coupled to a reference voltage.

[0005] In another exemplary aspect, an electronic device, including a device package, is disclosed. The device package includes a multi-layer substrate, a port hole through the multi-layer substrate and including a first opening on a first side of the multi-layer substrate and a second opening on a

second side of the multi-layer substrate, opposite to the first side. The device package further includes a device disposed on the first opening of the port hole and having a first voltage, an integrated circuit coupled to the device, and a spark arrestor including a first conductive layer of the multi-layer substrate disposed inside the port hole and coupled to a reference voltage.

[0006] In another exemplary aspect, a method of forming a package is disclosed. The method includes forming a multi-layer substrate and forming a port hole through the multi-layer substrate and including a first opening on a first side of the multi-layer substrate and a second opening on a second side of the multi-layer substrate opposite to the first side. The method further includes disposing a device having a first voltage on a first opening of the port hole and forming a spark arrestor inside the port hole, including a first conductive layer of the multi-layer substrate coupled to a reference voltage.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1A is a perspective view from one side of a conventional device package including a multi-layer substrate that may be employed in an electronic device, such as a smartphone, and includes a port hole through the multi-layer substrate through which a device disposed on the port hole on the other side of the multi-layer substrate can communicate with the environment;

[0008] FIG. 1B is a cross-sectional side view of the multi-layer substrate illustrated in FIG. 1A with the device disposed over the port hole on one side to show that a size of the opening of the port hole remains constant through the multi-layer substrate;

[0009] FIG. 1C is a perspective view of the device disposed on the port hole on one side of the multi-layer substrate in FIG. 1A for communicative access to the environment on the other side;

[0010] FIG. 2A is a perspective view of one side of an exemplary device package, including a multi-layer substrate that may be employed in an electronic device, such as a smartphone, and includes a port hole including a spark arrest shoulder to protect a device disposed on the port hole on the other side of the multi-layer substrate from sparks attracted to the device from the environment;

[0011] FIG. 2B is a cross-sectional side view of the multi-layer substrate illustrated in FIG. 2A, showing the device disposed over the port hole on one side and showing a first opening size of the port hole in a first substrate layer and a smaller opening size in another layer to provide a shoulder on which a spark arrestor is formed;

[0012] FIG. 3 is a flowchart of a method of forming a device package comprising a port hole in a multi-layer substrate, including a spark arrestor shoulder to protect the device disposed over the port hole;

[0013] FIG. 4 is an illustration of an example of a package substrate including contacts for coupling to the multi-layer substrate in FIGS. 2A-2B and a port hole axially aligned with the port hole in the multi-layer substrate to provide the device access to the environment through the package;

[0014] FIG. 5 is an illustration of a side view of an electronic device including a device package including the package substrate and multi-layer substrate in FIG. 4 and a device disposed on the port hole on the multi-layer substrate to communicate with the environment outside the electronic device;

[0015] FIG. 6A and 6B are a perspective view and a cross-sectional side view, respectively, of a second example of an exemplary device package, including a multi-layer substrate that may be employed in an electronic device, such as a smartphone, and including a port hole divided into two sections, with the spark arrestor also employed on the divider to improve the ability to attract sparks entering the port hole;

[0016] FIG. 7 is a block diagram of an exemplary wireless communication device that includes an exemplary device package including a multi-layer substrate with a port hole for environmental access and a spark arrestor shoulder to protect a device disposed on the port hole from an electric spark, as shown in FIGS. 2A, 2B, and 4; and

[0017] FIG. 8 is a block diagram of an exemplary processor-based system in an electronic device that may be included on an exemplary device package, including a multi-layer substrate with a port hole for environmental access and a spark arrest shoulder to protect a device disposed on the port hole from electric spark, as shown in FIGS. 2A, 2B, and 4.

#### DETAILED DESCRIPTION

[0018] Several exemplary aspects of the present disclosure are described in reference to the drawing figures. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects.

[0019] Aspects disclosed in the detailed description include electronic devices including device packages having a port hole with a spark arrest shoulder. Related methods of forming a device package, including a spark arrest shoulder, are also disclosed. The electronic device may include a device that receives information communicated in signals from the environment of the user outside of the device, which may include acoustic signals, radio signals, or optical signals, for example. The device is mounted on a first side of a multi-layer substrate as part of a device package. To provide communicative access to the environment, the device is mounted on a first opening of a port hole that extends through the multi-layer substrate from a second side, opposite to the first side, so that a signal from the environment may pass through the port hole to reach the device. In an exemplary aspect, the package includes a spark arrestor disposed inside the port hole and coupled to a reference voltage. The spark arrestor is provided to attract sparks to protect the device from damage due to electric sparks entering from the environment. In some examples, the port hole includes a second opening in a first substrate layer on the second side of the multi-layer substrate. In such examples, the second opening may be wider than a third opening in a second substrate layer, forming a shoulder region inside the port hole that is exposed through the second opening. The spark arrestor may include a conductive layer on the shoulder region.

[0020] FIG. 1A is a perspective view of a conventional device package **100** that may be employed in an electronic device (not shown), such as a smartphone. The device package **100** includes a multi-layer substrate **102** and a device **104** disposed on a first side **S1** of the multi-layer substrate **102**. The following description of the device package **100** also refers to features shown more clearly in FIG. 1B. The multi-layer substrate **102** includes a port hole **106** with the device **104** disposed on the port hole **106** on the first side **S1** of the multi-layer substrate **102** to allow the device **104** to communicate through the port hole **106** with the environment on a second side **S2** of the multi-layer substrate **102**. The device **104** may include an electrical circuit **108** coupled to a power source, such as a battery (not shown), to receive a supply voltage  $V_{sub,DD}$  and a reference voltage  $V_{sub,SS}$ . The device **104** may also be a mechanical or electromechanical device having a voltage  $V_{sub,104}$  independent of the supply voltage  $V_{sub,DD}$  and a reference voltage  $V_{sub,SS}$ .

[0021] Some examples of the device **104** may include micro-electromechanical (MEMS) devices configured to detect acoustic variations (e.g., a microphone) or air pressure. Other examples of the device **104** may include an audio speaker or an optical or wireless transmitter or receiver. Other examples may include integrated circuits (ICs). Receiving or transmitting any of acoustic signals, air pressure, optical signals, infrared, ultraviolet, wireless etc., are examples of the device **104** communicating with the environment on the second side **S2** through the port hole **106**. The device **104** may have one surface **110** configured for the communication, and the surface **110** is disposed on the first side **S1** of the multi-layer substrate **102** on (e.g., over) an opening **112(J)** in the port hole **106**. The port hole **106** may improve communication between the surface **110** and the environment.

[0022] The multi-layer substrate **102** in FIG. **1A** includes electrical contacts **114A-114D**, which may provide the supply voltage  $V_{sub.DD}$  and the reference voltage  $V_{sub.SS}$  from an external circuit. The electrical contacts **114A-114D** may also provide control and/or data signals to or from the electrical circuit **108** in the device **104**. The electrical contacts **114A-114D** are merely representative as there may be any number of contacts provided as needed for functions of the device package **100**. In the example in FIG. **1A**, the device **104** is an acoustic sensor. Thus, the device package **100** also includes a seal **116** provided on the surface of the second side **S2** between the multi-layer substrate **102** and a package substrate (not shown) to provide an acoustic seal around the port hole **106**.

[0023] FIG. **1B** is a cross-sectional side view of the multi-layer substrate **102** in FIG. **1A**. In FIG. **1B**, it can be seen that the multi-layer substrate **102** includes multiple conductive layers **118(1)-118(J)** alternating with dielectric layers **120(1)-120(J)**. The conductive layers **118(1)-118(J)** are formed of one or more conductive materials (e.g., metals), and the dielectric layers **120(1)-120(J)** may include or consist of dielectric and/or insulating materials (e.g., pre-preg). There may be any number  $J$  of the conductive layers **118(1)-118(J)** and a corresponding number  $J$  of the dielectric layers **120(1)-120(J)**. The conductive layers **118(1)-118(J)** may be patterned to form circuits that are interconnected by vias (not shown) through the dielectric layers **120(1)-120(J)**.

[0024] The port hole **106** is formed by the openings **112(1)-112(J)**, respectively, through the conductive layers **118(1)-118(J)** and the dielectric layers **120(1)-120(J)**. In the example in FIGS. **1A** and **1B**, it can be seen that each of the openings **112(1)-112(J)** have a same first width  $W1$  in a first direction ( $X$ -axis direction) and a same second width  $W2$  in a second direction ( $Y$ -axis direction). An axis  $A_{sub.100}$  of the port hole **106** extends orthogonally to the multi-layer substrate **102** through all of the openings **112(1)-112(J)**.

[0025] A problem encountered with the conventional device package **100** in FIGS. **1A** and **1B** is that sparks, such as from electrostatic discharge from a hand of a user, an external object, or the environment, may enter the port hole **106** and reach the electrical circuit **108** in the device **104**, causing damage to the device **104** that may render it inoperable. An electrostatic discharge may be attracted to the supply voltage  $V_{sub.DD}$ , the reference voltage  $V_{sub.SS}$  (not shown) coupled to the device **104**, or the independent voltage  $V_{sub.104}$  of the device.

[0026] FIG. **1C** is a perspective view showing the device **104** disposed on the port hole **106** on the first side **S1** of the multi-layer substrate **102** in the device package **100** in FIGS. **1A** and **1B**. The device **104** is disposed over the opening **112(J)** of the port hole **106** on the first side **S1**. Although not shown here, the first side **S1** may include other devices, such as integrated circuits and/or other electrical or electronic components that may be damaged by sudden bursts of electrical charge (i.e., sparks). One or more of these other devices may be electrically connected with the device **104** such that a spark striking the device **104** may be propagated to an electrically connected device, which can result in the other device(s) as well as the device **104**.

[0027] FIG. **2A** is a perspective view of one side of an exemplary device package **200**, including a multi-layer substrate **202** including a spark arrestor **204** that may be employed in an electronic device (not shown), such as a smartphone. The device package **200** includes a device **206** disposed on a port hole **208** on the first, back side **SI** of the multi-layer substrate **202**. The multi-layer substrate **202** includes a shoulder region **210** on which the spark arrestor **204** is provided to protect the device **206** from sparks attracted to or directed to the device **206** from outside the device package **200**.

[0028] In some aspects, the device package **200** includes features that correspond to the features of the device package **100** of FIGS. **1A-1C**. In the following description, reference may also be made to features shown more clearly in FIG. **2B**. For example, the device **206** may include an electrical circuit **212** coupled to a power source, such as a battery (not shown) to receive a supply voltage  $V_{sub.DD}$  and/or a package reference voltage  $V_{ss}$ . Examples of the device **206** include micro-electromechanical (MEMS) devices configured to detect acoustic variations (e.g., a microphone) or

air pressure, audio speakers or optical or wireless transmitters or receivers. Other examples of the device **206** may include mechanical devices, electromechanical devices, and electrical devices, such as integrated circuits (ICs). In this regard, the device **206** may have a first voltage  $V_{sub.206}$  that is different from the supply voltage  $V_{sub.DD}$  and the reference voltage  $V_{sub.SS}$  of the package **200**. The device **206** may have a first surface **216** configured for communication, and the device **206** is disposed on the first side **S1** of the multi-layer substrate **202** on (e.g., over) the first surface **216** facing a first opening **218(M)** (where  $M=4$  in this example) in the port hole **208**. The electrical circuit **212** may be coupled to other devices and/or components on the first side **S1** of the multi-layer substrate **202** by wires or by contacts between the first surface **216** and the first side **S1**. The multi-layer substrate **202** in FIG. 2A also includes electrical contacts **220A-220D** on the second side **S2**, which are representative of any number of contacts used to provide connections for signals and power between the device package **200** and an external circuit.

[0029] In the example in FIGS. 2A and 2B, the device **206** may be an acoustic sensor, such as a MEMS device. Thus, the device package **200** in this example includes an acoustic seal **222** provided on a first conductive layer **224(1)** on the second side **S2** of the multi-layer substrate **202** between the multi-layer substrate **202** and a package substrate (not shown) to provide an acoustic seal **222** around the port hole **208**. The acoustic seal **222** may also or alternatively be employed as an electrical contact. Thus, the acoustic seal **222** may be formed of a conductive material, such as metal, or a non-conductive material. In other examples of the device **206**, the device package **200** may not include the acoustic seal **222**.

[0030] FIG. 2B is a cross-sectional side view of the multi-layer substrate **202**. The multi-layer substrate **202** comprises alternating conductive layers **224(1)-224(M)** and dielectric layers **226(1)-226(M)**. In some examples, there may be an additional conductive layer (**224(M+1)**) (not shown) in the multi-layer substrate **202**. A single one of the conductive layers **224(1)-224(M)** paired with an adjacent one of the dielectric layers **226(1)-226(M)** is referred to herein as one of a plurality of substrate layers **228(1)-228(M)** of the multi-layer substrate **202**. The first substrate layer **228(1)** includes the first conductive layer **224(1)** and the first dielectric layer **226(1)**. The multi-layer substrate **202** comprises the first substrate layer **228(1)** on the second side **S2** and a second substrate layer **228(2)** between the first substrate layer **228(1)** and the first side **S1** of the multi-layer substrate **202**. The conductive layers **224(1)-224(M)** be patterned to form circuits that are interconnected by vias (not shown) through the dielectric layers **226(1)-226(M)**.

[0031] The port hole **208** in the multi-layer substrate **202** is formed by openings **218(1)-218(M)** through the substrate layers **228(1)-228(M)**. It should be understood that each of the openings **218(1)-218(M)** extends through one of the conductive layers **224(1)-224(M)** and an adjacent one of the dielectric layers **226(1)-226(M)**. In the exemplary device package **200**, a second opening **218(1)** of the port hole **208** through the first substrate layer **228(1)** is wider in a first direction (e.g., X-axis direction) than a third opening **218(2)** of the port hole **208** through the second substrate layer **228(2)**. That is, the second opening **218(1)** has a width  $WX1$  in the first direction and the third opening **218(2)** has a width  $WX2$  in the first direction, and  $WX2$  is smaller than  $WX1$ . In addition, the second opening **218(1)** has a larger width  $WY1$  in a second direction (Y-axis direction) than a width  $WY2$  in the second direction of the third opening **218(2)** through the second substrate layer **228(2)**. The second direction is orthogonal to the first direction. An axis  $A_{sub.200}$  of the port hole **208** extends in a third direction (Z-axis direction) that is orthogonal to both the first direction (X-axis) and the second direction (Y-axis). The axis  $A_{sub.200}$  of the port hole **208** is also an axis of all the openings **218(1)-218(M)**.

[0032] With the widths  $WX2$  and  $WY2$  of the second substrate layer **228(2)** being smaller than the corresponding widths  $WX1$  and  $WY1$  of the first substrate layer **228(1)**, the shoulder region **210** is a portion of the second substrate layer **228(2)** exposed through the second opening **218(1)** of the first substrate layer **228(1)** on the second side **S2** of the multi-layer substrate **202** in the third (Z-axis) direction. In the shoulder region **210**, the second substrate layer **228(2)** comprises a

conductive layer **224(2)** configured to be coupled to the reference voltage  $V_{sub.SS}$ . The conductive layer **224(2)** provides the spark arrestor **204**. In some examples, the conductive layer **224(2)** may include sections electrically isolated from each other and each coupled to different voltage potentials, such that the shoulder region **210** may be coupled to the reference voltage  $V_{sub.SS}$  or another voltage potential and other sections may be coupled to another voltage potential, which may include the supply voltage  $V_{sub.DD}$ . In some examples, the shoulder region **210** may be formed on the third substrate layer **228(3)**, for example, where a fourth opening **218(3)** is smaller (e.g., in at least one of the first and second directions) than the second opening **218(1)** and the third opening **218(2)**.

[0033] The conductive layer **224(2)** disposed in the shoulder region **210** may be a layer of conductive material (e.g., copper) disposed continuously around the entire perimeter of the third opening **218(2)**. In other examples, the conductive layer **224(2)** may be patterned (not shown) such that portions of the conductive layer **224(2)** are disposed on separate areas around the shoulder region **210**, such as around the perimeter of the third opening **218(2)**. Each of such portions is configured to couple to the reference voltage  $V_{sub.SS}$ , for example (not shown).

[0034] As noted, the conductive layer **224(2)** configured to couple to the reference voltage  $V_{sub.SS}$  and disposed on the exposed shoulder region **210** of the second substrate layer **228(2)** provides the spark arrestor **204**. That is, a spark arriving from any side or axially into the second opening **218(1)** is likely to pass close to the conductive layer **224(2)** because it is exposed on the shoulder region **210**. Such sparks may be electrically attracted to the voltage potential of the conductive layer **224(2)**, so the conductive layer **224(2)** provides an attractive path to the reference voltage  $V_{sub.SS}$  (e.g., ground). In this way, the forward progress of the electrical charge or spark is safely arrested before it reaches the first surface **216** of the device **206** disposed on the first opening **218(M)** on the first side **S1** of the multi-layer substrate **202**.

[0035] FIG. 3 is a flow chart of a method **300** of forming the device package **200**. The method comprises forming a multi-layer substrate **202** (block **302**); forming a port hole **208** through the multi-layer substrate **202** and comprising a first opening **218(M)** on a first side **S1** of the multi-layer substrate **202** and a second opening **218(1)** on a second side **S2** opposite to the first side **S1**, of the multi-layer substrate **202** (block **304**); and disposing a device **206** having a first voltage  $V_{sub.206}$  on the first opening **218(M)** (block **306**). The method also includes forming a spark arrestor **204** inside the port hole **208** and comprising a first conductive layer **224(2)** of the multi-layer substrate **202** coupled to a reference voltage  $V_{sub.SS}$  (block **308**).

[0036] FIG. 4 is a perspective view of elements of a package **400** detached from each other to more clearly show respective contacts for electrically coupling the elements, which include a device package **402** and a package substrate **404**. The device package **402** may be the device package **200** in FIGS. 2A and 2B and will not be described in further detail here. The package substrate **404** includes a port hole **406**. The package substrate **404** includes contacts **410A-410D** configured to align with contacts **412A-412D** of the device package **402**. The contacts **410A-410D** are merely representative of a number of contacts that may be provided between the device package **402** and the package substrate **404** to convey electrical power or signals between the device package **402** and an external circuit (not shown).

[0037] The package substrate **404** and the device package **402** are oriented such that an axis  $A_{sub.404}$  of the port hole **406** in the package substrate **404** is aligned (e.g., collinear) with an axis  $A_{sub.400}$  of the device package **402**. In this manner, a device **414** disposed on a port hole **416** in the device package **402** continues to have communicative access to the environment after the device package **402** is coupled to the package substrate **404**. The device package **402** is provided with a spark arrestor **418** on a shoulder region **420** to attract sparks entering the package **400** through the port hole **406** in the package substrate **404** to protect the device **414**.

[0038] FIG. 5 is an illustration of a side view of an electronic device **500**, including a package **502**, which may be the package **400** in FIG. 4. The package **502** includes a device package **504** coupled

to a package substrate **506**, corresponding to the device package **402** and the package substrate **404** in FIG. **4**, through contacts **507**. The device package **504** includes a multi-layer substrate **508**, which may be the multi-layer substrate **202** in FIGS. **2A** and **2B** and includes a device **510** disposed on a port hole **512** through the multi-layer substrate **508**. The package substrate **506** also includes a port hole **514** aligned with the port hole **512**. A seal **515** is disposed around the port hole **512** between the device package **504** and the package substrate **506**. The package **502** is disposed in a housing **516** that includes a port **518** aligned with the port holes **512** and **514** to provide communicative access to the environment for the device **510**. The port **518** may be an opening, a screen, a perforated surface, a membrane, a window or lens, etc., depending on the device **510**. In some examples, the port **518** provides a direct line-of-sight for the device **510** to communicate with the environment outside the electronic device **500**.

[0039] The multi-layer substrate **508** includes a shoulder region **520** including a conductive layer **522** providing a spark arrestor **524**, as discussed above, to protect the device **510** from sparks (e.g., electrostatic charges) that may enter the device **500** through the port **518**. The device **510** may be in electrical contact with other devices or components **526** through wires **528** or wire traces (not shown) on the first side **S1** of the multi-layer substrate **508**. The device **510** may include electrical circuits or provide electrical signals having a voltage  $V_{sub.510}$  while the component **526** includes circuits at a different voltage  $V_{sub.526}$ , and the spark arrestor **524** may receive the package reference voltage  $V_{ss}$ . Such devices or components **526** would also be protected from damage due to sparks by the spark arrestor **524**. FIGS. **6A** and **6B** are a perspective view and a cross-sectional view of another example of a multi-layer substrate **602** including a spark arrestor **604** in an electronic device (not shown), such as a smartphone. A device package **600** corresponds in many aspects to the device package **200** in FIGS. **2A** and **2B**. The device package includes a device **606** disposed on a port hole **608** on the first, back side **S1** of the multi-layer substrate **602**. The multi-layer substrate **602** includes a shoulder region **610** on which the spark arrestor **604** is provided to protect the device **606** from sparks that may be attracted or directed to the device **606** from outside the device package **600**.

[0040] In this example, the port hole **608** includes a divider **612** extending between two sections **614A** and **614B** of the port hole **608**. To improve the capability of the spark arrestor **604** to protect the device **606**, the spark arrestor **604** disposed on the shoulder region **610** extends onto the divider **612**. In this regard, spark protection is provided across a middle region of the port hole **608** rather than on the perimeter alone.

[0041] Electronic devices, according to any aspects disclosed herein, may be provided in or integrated into any processor-based device. Examples, without limitation, include a set-top box, an entertainment unit, a navigation device, a communications device, a fixed location data unit, a mobile location data unit, a global positioning system (GPS) device, a mobile phone, a cellular phone, a smartphone, a session initiation protocol (SIP) phone, a tablet, a phablet, a server, a computer, a portable computer, a mobile computing device, laptop computer, a wearable computing device (e.g., a smartwatch, a health or fitness tracker, eyewear, etc.), a desktop computer, a personal digital assistant (PDA), a monitor, a computer monitor, a television, a tuner, a radio, a satellite radio, a music player, a digital music player, a portable music player, a digital video player, a video player, a digital video disc (DVD) player, a portable digital video player, an automobile, a vehicle component, an avionics system, a drone, and a multicopter.

[0042] In this regard, FIG. **7** illustrates a block diagram of an exemplary wireless communications device **700** that includes radio frequency (RF) components formed from one or more ICs **702**, wherein the communications device **700** may be the electronic device **500** in FIG. **5** including the device package **504** and/or package **502**, which correspond to the device package **200** and package **400** illustrated in FIGS. **2A**, **2B**, and **4**, including a device disposed on a port hole in a multi-layer substrate for communicative access to the environment and a shoulder region comprising a spark arrestor formed on a second substrate layer having a smaller opening than a first substrate layer to



protect the device from damage caused by sparks. The wireless communications device **700** may include or be provided as examples in any of the above-referenced devices. As shown in FIG. 7, the wireless communications device **700** includes a transceiver **704** and a data processor **706**. The data processor **706** may include a memory to store data and program codes. The transceiver **704** includes a transmitter **708** and a receiver **710**, which support bi-directional communications. In general, the wireless communications device **700** may include any number of transmitters **708** and/or receivers **710** for any number of communication systems and frequency bands. All or a portion of the transceiver **704** may be implemented on one or more analog ICs, RF ICs (RFICs), mixed-signal ICs, etc.

[0043] The transmitter **708** or the receiver **710** may be implemented with a super-heterodyne or direct-conversion architecture. In the super-heterodyne architecture, a signal is frequency-converted between RF and baseband in multiple stages, e.g., from RF to an intermediate frequency (IF) in one stage and then from IF to baseband in another stage. In the direct-conversion architecture, a signal is frequency-converted between RF and baseband in one stage. The super-heterodyne and direct-conversion architectures may use different circuit blocks and/or have different requirements. In the wireless communications device **700** in FIG. 7, the transmitter **708** and the receiver **710** are implemented with the direct-conversion architecture.

[0044] In the transmit path, the data processor **706** processes data to be transmitted and provides I and Q analog output signals to the transmitter **708**. In the exemplary wireless communications device **700**, the data processor **706** includes digital-to-analog converters (DACs) **712(1)**, **712(2)** for converting digital signals generated by the data processor **706** into I and Q analog output signals, e.g., I and Q output currents, for further processing.

[0045] Within the transmitter **708**, lowpass filters **714(1)**, **714(2)** filter the I and Q analog output signals, respectively, to remove undesired signals caused by the prior digital-to-analog conversion. Amplifiers (AMPs) **716(1)**, **716(2)** amplify the signals from the lowpass filters **714(1)**, **714(2)**, respectively, and provide I and Q baseband signals. An upconverter **718** upconverts the I and Q baseband signals with I and Q transmit (TX) local oscillator (LO) signals from a TX LO signal generator **722** through mixers **720(1)**, **720(2)** to provide an upconverted signal **724**. A filter **726** filters the upconverted signal **724** to remove undesired signals caused by the frequency upconversion and noise in a receive frequency band. A power amplifier (PA) **728** amplifies the upconverted signal **724** from the filter **726** to obtain the desired output power level and provides a transmit RF signal. The transmit RF signal is routed through a duplexer or switch **730** and transmitted via an antenna **732**.

[0046] In the receive path, the antenna **732** receives signals transmitted by base stations and provides a received RF signal, which is routed through the duplexer or switch **730** and provided to a low noise amplifier (LNA) **734**. The duplexer or switch **730** is designed to operate with a specific receive (RX)-to-TX duplexer frequency separation, such that RX signals are isolated from TX signals. The received RF signal is amplified by the LNA **734** and filtered by a filter **736** to obtain a desired RF input signal. Downconversion mixers **738(1)**, **738(2)** mix the output of the filter **736** with I and Q RX LO signals (i.e., LO\_I and LO\_Q) from an RX LO signal generator **740** to generate I and Q baseband signals. The I and Q baseband signals are amplified by AMPs **742(1)**, **742(2)** and further filtered by lowpass filters **744(1)**, **744(2)** to obtain I and Q analog input signals, which are provided to the data processor **706**. In this example, the data processor **706** includes analog-to-digital converters (ADCs) **746(1)**, **746(2)** for converting the analog input signals into digital signals to be further processed by the data processor **706**.

[0047] In the wireless communications device **700** of FIG. 7, the TX LO signal generator **722** generates the I and Q TX LO signals used for frequency upconversion, while the RX LO signal generator **740** generates the I and Q RX LO signals used for frequency downconversion. Each LO signal is a periodic signal with a particular fundamental frequency. A TX phase-locked loop (PLL) circuit **748** receives timing information from the data processor **706** and generates a control signal

used to adjust the frequency and/or phase of the TX LO signals from the TX LO signal generator **722**. Similarly, an RX PLL circuit **750** receives timing information from the data processor **706** and generates a control signal used to adjust the frequency and/or phase of the RX LO signals from the RX LO signal generator **740**.

[0048] FIG. **8** illustrates a block diagram of an example of a processor-based system **800** that may be implemented in the electronic device **500** in FIG. **5** including the device package **504** and/or package **502**, which correspond to the device package **200** and package **400** illustrated in FIGS. **2A**, **2B**, and **4**, including a device disposed on a port hole in a multi-layer substrate for communicative access to the environment and a shoulder region comprising a spark arrestor formed on a second substrate layer having a smaller opening than a first substrate layer to protect the device from damage caused by sparks. In this example, the processor-based system **800** includes a processor **802** that includes an IC **804** including one or more central processor units (CPUs) **808**, which may also be referred to as CPU or processor cores, each including one or more processors **810**. The CPU(s) **808** may have cache memory **812** coupled to the processor(s) **802** for rapid access to temporarily stored data. The CPU(s) **808** is coupled to a system bus **814** and can intercouple master and slave devices included in the processor-based system **800**. As is well known, the CPU(s) **808** communicates with these other devices by exchanging address, control, and data information over the system bus **814**. For example, the CPU(s) **808** can communicate bus transaction requests to a memory controller **816** as an example of a slave device. Although not illustrated in FIG. **8**, multiple system buses **814** could be provided wherein each system bus **814** constitutes a different fabric.

[0049] Other master and slave devices can be connected to the system bus **814**. As illustrated in FIG. **8**, these devices can include a memory system **820** that includes the memory controller **816** and one or more memory arrays **818**, one or more input devices **822**, one or more output devices **824**, one or more network interface devices **826**, and one or more display controllers **828**, as examples. The input device(s) **822** can include any type of input device, including, but not limited to, input keys, switches, voice processors, etc. The output device(s) **824** can include any type of output device, including, but not limited to, audio, video, other visual indicators, etc. The network interface device(s) **826** can be any device configured to allow an exchange of data to and from a network **830**. The network **830** can be any type of network, including, but not limited to, a wired or wireless network, a private or public network, a local area network (LAN), a wireless local area network (WLAN), a wide area network (WAN), a BLUETOOTH™ network, and the Internet. The network interface device(s) **826** can be configured to support any type of communications protocol desired.

[0050] The CPU(s) **808** may also be configured to access the display controller(s) **828** over the system bus **814** to control information sent to one or more displays **832**. The display controller(s) **828** sends information to the display(s) **832** to be displayed via one or more video processors **834**, which process the information to be displayed into a format suitable for the display(s) **832**. The display(s) **832** can include any type of display, including, but not limited to, a cathode ray tube (CRT), a liquid crystal display (LCD), a plasma display, or a light-emitting diode (LED) display, etc.

[0051] Those of skill in the art will further appreciate that the various illustrative logical blocks, modules, circuits, and algorithms described in connection with the aspects disclosed herein may be implemented as electronic hardware, instructions stored in memory or in another computer-readable medium wherein any such instructions are executed by a processor or other processing device, or combinations of both. As examples, the devices and components described herein may be employed in any circuit, hardware component, integrated circuit (IC), or IC chip. Memory disclosed herein may be any type and size of memory and may be configured to store any desired information. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality.

How such functionality is implemented depends upon the particular application, design choices, and/or design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure. [0052] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0053] The aspects disclosed herein may be embodied in hardware and in instructions that are stored in hardware and may reside, for example, in Random Access Memory (RAM), flash memory, Read Only Memory (ROM), Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer-readable medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a remote station. Alternatively, the processor and the storage medium may reside as discrete components in a remote station, base station, or server.

[0054] It is also noted that the operational steps described in any of the exemplary aspects herein are described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications, as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using various technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0055] The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the spirit or scope of the disclosure. Thus, the disclosure is not intended to be limited to the examples and designs described herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

[0056] Implementation examples are described in the following numbered clauses:

[0057] 1. A package comprising: [0058] a multi-layer substrate; [0059] a port hole through the multi-layer substrate comprising a first opening on a first side of the multi-layer substrate and a second opening on a second side of the multi-layer substrate, opposite to the first side; and [0060] a device disposed on the first opening of the port hole and having a first voltage; and [0061] a spark arrestor comprising a first conductive layer of the multi-layer substrate disposed inside the port hole and coupled to a reference voltage.

[0062] 2. The package of clause 1, wherein: [0063] the multi-layer substrate comprises: [0064] a

first substrate layer on the second side of the multi-layer substrate; and [0065] a second substrate layer between the first substrate layer and the first side of the multi-layer substrate; [0066] the second opening of the port hole through the first substrate layer is wider in a first direction than a third opening of the port hole through the second substrate layer; [0067] a shoulder region of the second substrate layer is exposed through the second opening from the second side of the multi-layer substrate; and [0068] the first conductive layer is disposed in the shoulder region of the second substrate layer.

[0069] 3. The package of clause 2, wherein: [0070] the second substrate layer further comprises a first dielectric layer between the first conductive layer and the first side of the multi-layer substrate; and [0071] the first substrate layer further comprises a second conductive layer and a second dielectric layer between the second conductive layer and the second substrate layer.

[0072] 4. The package of clause 3, the multi-layer substrate comprising at least a third substrate layer comprising a third conductive layer and a third dielectric layer between the second substrate layer and the first side of the multi-layer substrate.

[0073] 5. The package of clause 2, wherein: [0074] the second opening of the port hole is wider in a second direction than the third opening of the port hole; and [0075] the second direction is orthogonal to the first direction.

[0076] 6. The package of clause 5, wherein an axis of the port hole is orthogonal to the first direction and the second direction.

[0077] 7. The package of clause 2, wherein each of the second opening and the third opening of the port hole comprises one of a round opening and a polygonal opening.

[0078] 8. The package of clause 2, further comprising: [0079] a package substrate coupled to the second side of the multi-layer substrate; and a hole through the package substrate; [0080] wherein a longitudinal axis of the hole through the package substrate extends along a longitudinal axis of the hole through the multi-layer substrate.

[0081] 9. The package of clause 8, further comprising: [0082] an acoustic seal disposed around the second opening between the first substrate layer and the package substrate; [0083] wherein the device comprises an acoustic device.

[0084] 10. The package of clause 9, wherein the acoustic seal comprises a metal layer coupled to the reference voltage.

[0085] 11. The package of clause 1, wherein the device comprises an electromechanical device.

[0086] 12. The package of clause 1, wherein the device comprises an environmental sensor.

[0087] 13. The package of clause 2, wherein the shoulder region extends around an entire perimeter of the third opening.

[0088] 14. The package of clause 2, wherein the first conductive layer extends continuously around an entire perimeter of the third opening.

[0089] 15. The package of clause 2, wherein the first conductive layer comprises a plurality of portions disposed around the shoulder region and each of the plurality of portions is coupled to the reference voltage.

[0090] 16. The package of clause 1 integrated into a device selected from the group consisting of: a set-top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smartphone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; an avionics system; a drone; and a multicopter.

[0091] 17. An electronic device comprising: [0092] a device package comprising: [0093] a multi-layer substrate; [0094] a port hole through the multi-layer substrate and comprising a first opening

on a first side of the multi-layer substrate and a second opening on a second side of the multi-layer substrate, opposite to the first side; [0095] a device disposed on the first opening of the port hole and having a first voltage; [0096] an integrated circuit coupled to the device; and [0097] a spark arrestor comprising a first conductive layer of the multi-layer substrate disposed inside the port hole and coupled to a reference voltage.

[0098] 18. The electronic device of clause 17, wherein: [0099] the multi-layer substrate comprises: [0100] a first substrate layer on the second side of the multi-layer substrate; and [0101] a second substrate layer between the first substrate layer and the first side of the multi-layer substrate; [0102] a second opening of the port hole through the first substrate layer is wider in a first direction than a third opening of the port hole through the second substrate layer; [0103] a shoulder region of the second substrate layer is exposed through the second opening from the second side of the multi-layer substrate; and [0104] the second substrate layer comprises a first conductive layer in the shoulder region coupled to the reference voltage.

[0105] 19. The electronic device of clause 18, further comprising: [0106] a package substrate coupled to the second side of the multi-layer substrate; and [0107] a hole through the package substrate; [0108] wherein a longitudinal axis of the hole through the package substrate extends along a longitudinal axis of the hole through the multi-layer substrate.

[0109] 20. A method of forming a package comprising: [0110] forming a multi-layer substrate; [0111] forming a port hole through the multi-layer substrate and comprising a first opening on a first side of the multi-layer substrate and a second opening on a second side of the multi-layer substrate, opposite to the first side; [0112] disposing a device having a first voltage on the first opening of the port hole; and [0113] forming a spark arrestor inside the port hole comprising a first conductive layer of the multi-layer substrate coupled to a reference voltage.

## Claims

**1.** A package comprising: a multi-layer substrate; a port hole through the multi-layer substrate and comprising a first opening on a first side of the multi-layer substrate and a second opening on a second side of the multi-layer substrate, opposite to the first side; and a device disposed on the first opening of the port hole and having a first voltage; and a spark arrestor comprising a first conductive layer of the multi-layer substrate disposed inside the port hole and coupled to a reference voltage.

**2.** The package of claim 1, wherein: the multi-layer substrate comprises: a first substrate layer on the second side of the multi-layer substrate; and a second substrate layer between the first substrate layer and the first side of the multi-layer substrate; the second opening of the port hole through the first substrate layer is wider in a first direction than a third opening of the port hole through the second substrate layer; a shoulder region of the second substrate layer is exposed through the second opening from the second side of the multi-layer substrate; and the first conductive layer is disposed in the shoulder region of the second substrate layer.

**3.** The package of claim 2, wherein: the second substrate layer further comprises a first dielectric layer between the first conductive layer and the first side of the multi-layer substrate; and the first substrate layer further comprises a second conductive layer and a second dielectric layer between the second conductive layer and the second substrate layer.

**4.** The package of claim 3, the multi-layer substrate comprising at least a third substrate layer comprising a third conductive layer and a third dielectric layer between the second substrate layer and the first side of the multi-layer substrate.

**5.** The package of claim 2, wherein: the second opening of the port hole is wider in a second direction than the third opening of the port hole; and the second direction is orthogonal to the first direction.

**6.** The package of claim 5, wherein an axis of the port hole is orthogonal to the first direction and

the second direction.

**7.** The package of claim 2, wherein each of the second opening and the third opening of the port hole comprises one of a round opening and a polygonal opening.

**8.** The package of claim 2, further comprising: a package substrate coupled to the second side of the multi-layer substrate; and a hole through the package substrate; wherein a longitudinal axis of the hole through the package substrate extends along a longitudinal axis of the hole through the multi-layer substrate.

**9.** The package of claim 8, further comprising: an acoustic seal disposed around the second opening between the first substrate layer and the package substrate; wherein the device comprises an acoustic device.

**10.** The package of claim 9, wherein the acoustic seal comprises a metal layer coupled to the reference voltage.

**11.** The package of claim 1, wherein the device comprises an electromechanical device.

**12.** The package of claim 1, wherein the device comprises an environmental sensor.

**13.** The package of claim 2, wherein the shoulder region extends around an entire perimeter of the third opening.

**14.** The package of claim 2, wherein the first conductive layer extends continuously around an entire perimeter of the third opening.

**15.** The package of claim 2, wherein the first conductive layer comprises a plurality of portions disposed around the shoulder region and each of the plurality of portions is coupled to the reference voltage.

**16.** The package of claim 1 integrated into a device selected from the group consisting of: a set-top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smartphone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; an avionics system; a drone; and a multicopter.

**17.** An electronic device comprising: a device package comprising: a multi-layer substrate; a port hole through the multi-layer substrate and comprising a first opening on a first side of the multi-layer substrate and a second opening on a second side of the multi-layer substrate, opposite to the first side; a device disposed on the first opening of the port hole and having a first voltage; an integrated circuit coupled to the device; and a spark arrestor comprising a first conductive layer of the multi-layer substrate disposed inside the port hole and coupled to a reference voltage.

**18.** The electronic device of claim 17, wherein: the multi-layer substrate comprises: a first substrate layer on the second side of the multi-layer substrate; and a second substrate layer between the first substrate layer and the first side of the multi-layer substrate; a second opening of the port hole through the first substrate layer is wider in a first direction than a third opening of the port hole through the second substrate layer; a shoulder region of the second substrate layer is exposed through the second opening from the second side of the multi-layer substrate; and the second substrate layer comprises a first conductive layer in the shoulder region coupled to the reference voltage.

**19.** The electronic device of claim 18, further comprising: a package substrate coupled to the second side of the multi-layer substrate; and a hole through the package substrate; wherein a longitudinal axis of the hole through the package substrate extends along a longitudinal axis of the hole through the multi-layer substrate.

**20.** A method of forming a package comprising: forming a multi-layer substrate; forming a port hole through the multi-layer substrate and comprising a first opening on a first side of the multi-

layer substrate and a second opening on a second side of the multi-layer substrate, opposite to the first side; disposing a device having a first voltage on the first opening of the port hole; and forming a spark arrestor inside the port hole comprising a first conductive layer of the multi-layer substrate coupled to a reference voltage.

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